

NANOOM LEE

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EMPLOYMENT

Horizon Postdoctoral Fellow · Johns Hopkins University · Baltimore, MD

Oct 2024 – Present

- Mentor: Prof. Marc Kamionkowski

EDUCATION

Ph.D., Physics · New York University, New York, NY

Jan 2025

Advisors: Profs. Yacine Ali-Haïmoud & Roman Scoccimarro

Thesis: Imprints of Fundamental Physics on the Primordial Cosmic Plasma

M.A., Physics · Stony Brook University, Stony Brook, NY

May 2018

B.S., Physics & Mathematics · Korea University, Seoul, South Korea

Feb 2016

First in class

RESEARCH INTERESTS

Data-driven cosmology, CMB, Cosmological recombination, Boltzmann code development, Extensions to Λ CDM (decaying dark matter), Hubble tension, Large-scale structure, Fisher forecasting, Statistical methods.

HONORS & AWARDS

- Newton International Fellowship — The Royal Society (reserve list) 2024
- Roman Galaxy Redshift Survey Postdoctoral Researcher — Caltech (declined) 2024
- Beus Prize Postdoctoral Fellowship — Arizona State University (finalist, 2nd place) 2024
- James Arthur Graduate Associate Fellowship — NYU 2020 – 2021, 2022 – 2023
- Balzan Cosmological Studies Program Award — Oxford / JHU 2022
- **Outstanding Graduate Student Instructor Award** — NYU 2019 – 2020
- **Korean Government Scholarship for Overseas Study** — Ministry of Education, South Korea 2016 – 2018
- First Rank Graduation Award — Korea University 2016
- Boheon Scholarship (Full tuition) — Korea University 2012 – 2013, 2015
- National Scholarship — Ministry of Education, South Korea 2012

PUBLICATIONS

11 first-authored papers (14 total) · h -index 7 · $\gtrsim$ 250 citations · Google Scholar · InspireHEP

[†] mentored students

First- & second-authored

1. A. Bencke[†], **N. Lee**, M. Kamionkowski “Constraints with CMB lensing on dark matter decays to massive decay products” arXiv:2606.14849 (under review)
2. **N. Lee** and T. Zhou[†] “What it takes to solve the Hubble tension through Modifications of Cosmological Recombination II: in light of ACT DR6 and DESI DR2” arXiv:2606.06495 (under review)
3. **N. Lee** “Compressed Gaussian likelihood for the Planck low- ℓ data” arXiv:2606.05166 (under review)
4. **N. Lee**, Y. Ali-Haïmoud, M. Kamionkowski “Efficient analytic approximation for small-scale non-cold relic perturbations” Phys. Rev. D 113, 043525 (2026)
5. **N. Lee**, J. L. Bernal, S. Günther, L. Ji, M. Kamionkowski “Rapid and accurate numerical evolution of linear cosmological perturbations with non-cold relics” Phys. Rev. D 112, 043529 (2025)
6. **N. Lee**, M. Braglia, Y. Ali-Haïmoud “What it takes to solve the Hubble tension through scale-dependent modifications to primordial power spectrum” Phys. Rev. D 112, 083506 (2025)
7. A. Eggemeier, **N. Lee**, R. Scoccimarro, et al. “Boosting galaxy clustering analyses with non-perturbative modelling of redshift-space distortions” Phys. Rev. D 112, 063532 (2025)
8. **N. Lee** and Y. Ali-Haïmoud “Magnetic field from primordial perturbations” Phys. Rev. D 109, 103536 (2024)
9. **N. Lee** and S. C. Hotinli “Probing light relics through cosmic dawn” Phys. Rev. D 109, 043502 (2024)

10. **N. Lee**, Y. Ali-Haïmoud, N. Schöneberg, V. Poulin “What it takes to solve the Hubble tension through modifications of cosmological recombination” Phys. Rev. Lett. 130, 161003 (2023)
11. **N. Lee**, S. C. Hotinli, M. Kamionkowski “Probing cosmic birefringence with Polarized Sunyaev Zel’dovich Tomography” Phys. Rev. D 106, 083518 (2022)
12. **N. Lee** and Y. Ali-Haïmoud “Probing small-scale baryon and dark matter isocurvature perturbations with cosmic microwave background anisotropies” Phys. Rev. D 104, 103509 (2021)
13. **N. Lee** and Y. Ali-Haïmoud “HYREC-2: a highly accurate sub-millisecond recombination code” Phys. Rev. D 102, 083517 (2020)

Nth-authored

1. A. Pezzotta et al. “Extending evolution mapping to massive neutrinos with COMET” Phys. Rev. D 112, 023520 (2025)

In preparation

1. **N. Lee**, A. Bencke[†], M. Kamionkowski “Cosmological evolution with decaying dark matter”

PUBLIC CODE

planck-gaussian-lowl	github.com/nanoomlee/planck-gaussian-lowl Compressed Gaussian likelihood for the Planck low- ℓ EE data Piecewise log-normal approximation enabling analytic Fisher matrix computation
CLASSIER	github.com/nanoomlee/CLASSIER CLASS Integral Equation Revision (a modified version of CLASS) Boltzmann hierarchy for non-cold relics replaced with integral-equation approach
HYREC-2	github.com/nanoomlee/HYREC-2 A highly accurate sub-millisecond cosmological recombination code Incorporated into CLASS and CAMB

TALKS & PRESENTATIONS

Oct 2025	CosmoLab, USC
Oct 2025	Cahill Cosmology Journal Club, Caltech (remote)
Oct 2025	COSMO-25, CMU
Jul 2025	Cosmology Summer Fest 2025, JHU, Baltimore
Jun 2025	Cosmology from Home (remote)
Jan 2024	Perimeter Institute, Waterloo (remote)
Apr 2023	PONT conference, Avignon
Apr 2023	LUPM, Montpellier
Jan 2023	AAS 241st Meeting (iPoster), Seattle
Jul 2022	Cosmology from Home (remote)
May 2022	Particle Astro/Cosmo Meeting Around NYC (PACMAN), CCA
Apr 2022	Brown Bag, New York University
May 2021	NYU-CCA × Data Science meeting

TECHNICAL SKILLS

Programming	Python (Pandas, NumPy, SciPy, scikit-learn), C, Mathematica, HPC/parallel computing, git, Linux
Research Tools	CLASS, CAMB, MontePython, MultiNest, emcee

MENTORSHIP

Anna Bencke	Ph.D. student, JHU (1 paper under review, 1 paper in preparation)	2025 –
Tianji (Jimmy) Zhou	Undergraduate, Haverford College → USC Ph.D program (1 paper under review)	2025 –

SERVICE

Referee: Nature Astronomy, Astronomy & Astrophysics, Journal of Cosmology and Astroparticle Physics

TEACHING

General Physics I Lab (undergraduate)
Electricity & Magnetism I (undergraduate)
Mathematical Physics (undergraduate)

Spring 2024
Fall 2019, Fall 2021
Spring 2019, Spring 2020

REFERENCES

Yacine Ali-Haïmoud

Ph.D. advisor

Associate Professor, Department of Physics, New York University

yah2@nyu.edu

Roman Scoccimarro

Ph.D. advisor (secondary)

Professor, Department of Physics, New York University

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Marc Kamionkowski

Postdoc mentor

William R. Kenan Jr. Professor, Department of Physics & Astronomy, JHU

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